

International Nurses and Housing in Ireland Analysis Report



ARE WE SENDING INTERNATIONALLY
RECRUITED NURSES TO THE PLACES THAT
NEED THEM MOST, ONLY TO FIND THEY
CANNOT AFFORD TO LIVE THERE?

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Data Analytics & Visualization



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Introduction

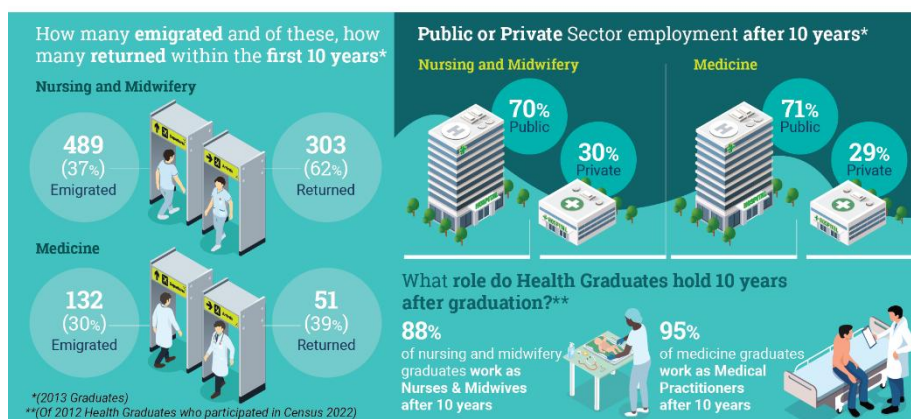
The Irish healthcare system is dependent on international nurses to meet growing demands: more than 40 per cent of doctors and 50 per cent of nurses in Ireland are foreign-trained. Ireland has one of the highest numbers of nurses per capita in Europe MCHALE, M. (2024), yet thousands emigrate overseas to work in Australia, Canada, the UK, and the Middle East. Among 2013 Irish-trained nursing and midwifery graduates, approximately 37 per cent emigrated within ten years of qualifying, of whom 62 per cent returned within that same period CSO (2023). To fill these workforce gaps, Ireland continues to recruit nurses from countries such as India, the Philippines, and parts of Africa.

Recruitment, however, is only half the challenge. Retention depends on whether these nurses can settle in Ireland, and housing is central to this. My prior experience in healthcare recruitment (2022 - 2024) and real estate (2024) gave me direct insight into the challenges internationally recruited nurses face upon arrival. I helped nurses and healthcare assistants, primarily from India, the Philippines, and parts of Africa, register for Nursing and Midwifery Board of Ireland (NMBI) examinations and secure work permits through the Department of Enterprise, Trade and Employment (DETE). These newly arrived nurses typically received just one month of temporary accommodation upon arrival, after which they were left to navigate Ireland's ongoing housing crisis independently.

When I moved into real estate in 2024, I witnessed firsthand the demand for rental accommodation and the chronic lack of affordable supply, particularly affecting lower-income migrants such as newly arrived nurses. While established nurses working in Ireland for several years had typically secured homes through purchase or long-term rental, newly arriving nurses frequently struggled to find safe, affordable accommodation close to their workplace often while earning on the lowest salary scale. Accessing a mortgage compounded this difficulty further: nurses cannot apply for a mortgage until they have at least six months of confirmed permanent employment, but permanent status itself cannot be granted until after a six-month probationary period. In practice, this means a minimum wait of one full year before any mortgage application is possible, assuming visa status and income allow it.



Health Graduate Outcomes 2023 Graduation Years 2010 - 2022



DATA DOMAIN

This report links nursing workforce data, hospital locations, housing market data, and county-level demographic and income data. Together, they show where nurses work, where hospitals are located, and where housing is least affordable.

AIMS & OBJECTIVES

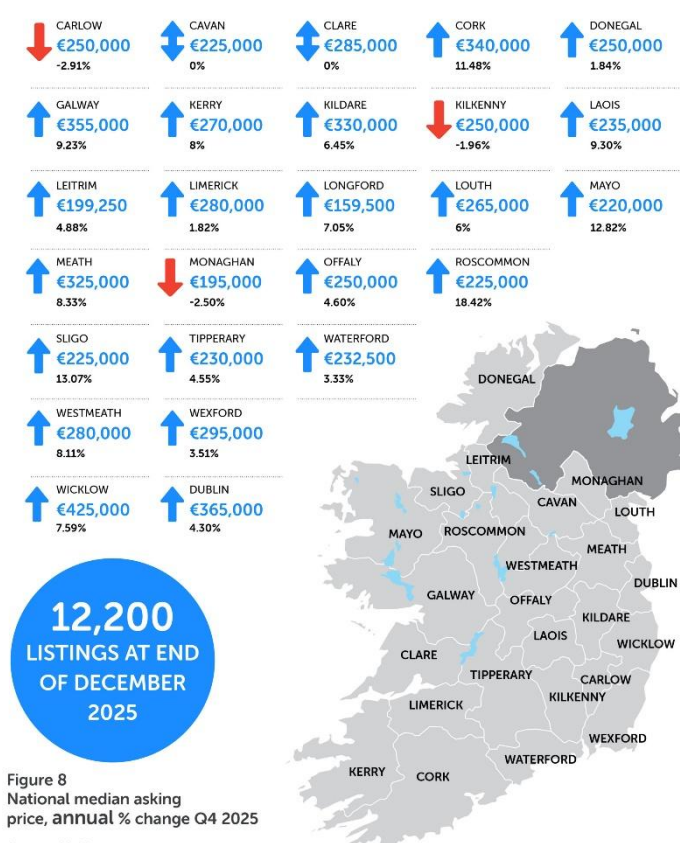
Aim:

To investigate the relationship between the geographic distribution of internationally recruited nurses in Ireland and housing affordability and supply across the counties where those nurses are most needed.

Objectives:

1. Extract and clean nursing data from the NMBI State of the Register 2025, focusing on the geographic distribution of all nurses by county.
2. Compile and process hospital location data from the GeoHive dataset to identify the location of healthcare facilities in Ireland.
3. Compile housing market data from the CSO, RTB and PPR, to include mean purchase prices and rental costs per county.
4. Calculate key metrics including nurse concentration rates, estimated international nurse numbers, average nurse salary, and housing affordability ratios for each county.
5. Analyse the relationship between international nurse distribution, housing affordability and hospital placement.
6. Visualise findings using charts and maps to tell a clear story.

Note: When calculating average house prices, I will use the actual sale prices recorded on the Property Price Register, not the advertised asking prices or the national median asking price. This is important because most homes in Ireland are currently selling at 7.4% above asking price (MACCOILLE, C. ET AL. (2026) Using advertised or median asking prices would make houses seem more affordable than they really are.



DATA SOURCES

The NMBI register is the main workforce source because registration is mandatory, so it gives a full census rather than a sample. GeoHive hospital data provides hospital locations and coordinates for mapping. Housing data comes from the Property Price Register and RTB Rent Index, giving actual sale prices and county rents. CSO population data and HSE salary scales provide the denominator for affordability ratios.

Tables extracted from NMBI and their relevance to the report:

Fig 1. Top 20 nationalities on the NMBI Register.

Identifies origin countries of internationally recruited nurses

	Country	Registered		Country	Registered
1	Ireland	54,436	11	Spain	328
2	India	20,419	12	Portugal	306
3	Philippines	7,324	13	Italy	259
4	United Kingdom	3,151	14	Croatia	220
5	Nigeria	973	15	Australia	175
6	Zimbabwe	856	16	Germany	154
7	Poland	677	17	Lithuania	145
8	Romania	641	18	South Africa	131
9	Ghana	414	19	Kenya	90
10	United States	346	20	Canada	81

*There are 117 nationalities on the NMBI Register. Nationality is captured at the time of registration and it is the responsibility of the registrant to update NMBI of any changes.

(NMBI STATE OF THE REGISTER, 2025)

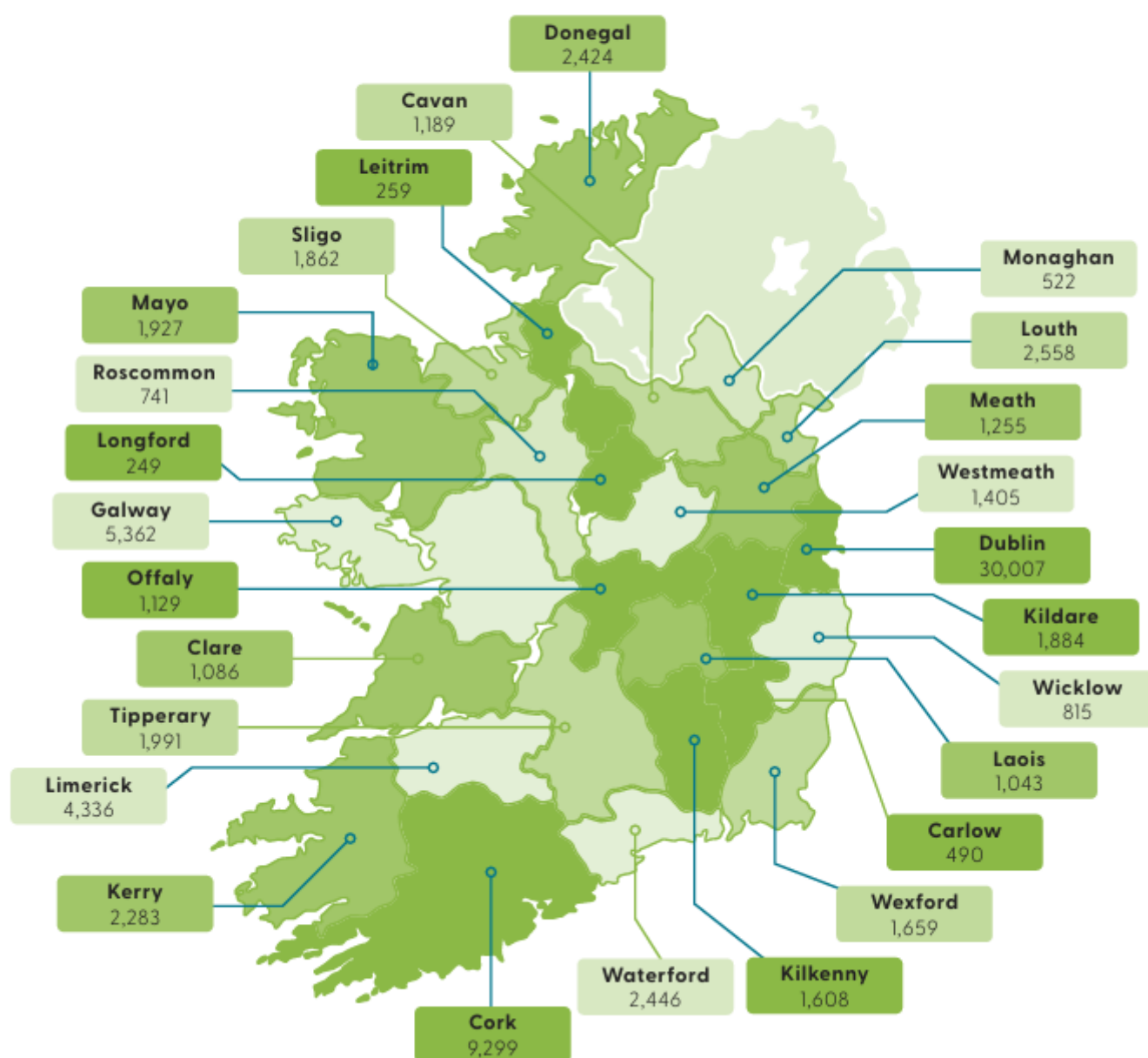
Fig 2. Country of qualification of registered, practising and patient-facing registrants.

Distinguishes international from domestic recruits at national level

	Registered	% Registered	Practising	% Practising	Patient-facing	% Patient-facing
EU educated	2,851	3%	2,578	3%	2,393	3%
Irish educated	45,640	49%	41,938	48%	37,041	47%
Non-EU educated	43,894	48%	42,432	49%	39,760	50%

(NMBI STATE OF THE REGISTER, 2025)

Fig 3. Geographical workplace of all registrants practising in Ireland.
This helps map the nursing workforce at county level.



(NMBI STATE OF THE REGISTER, 2025)

ETHICAL AND LICENCING RESTRICTIONS

All datasets selected for this project are publicly available and under open data licences that are free to share and adapt. All data is aggregated and anonymous, so no ethical approval is needed.

Data Source	Licence Type	Ethical Considerations
CSO Population Data	CSO Open Data Licence	Aggregate census data only, no individuals identified
NMBI State of the Register 2025	Public Report	Data aggregated to county level, no individual identifiers
GeoHive (HSE Hospitals)	CC BY 4.0 Open Data Licence	No personal data, facility locations only
Property Services Regulatory Authority (PSRA)	PSRA Open Data Licence	Anonymised statistical data only
Residential Property Price Register	CSO Open Data Licence	
RTB Average Monthly Rent Report 2025 Q1-Q3	CSO Open Data Licence	Aggregated regional data only
HSE Salary Scales	Public information	Publicly available, no personal data involved

STORING DATA

Data will be stored on Google Drive to ensure both security and accessibility:

All files will be stored in the project folder CA2_L00203071/. Within the project folder, data will be organised into two subfolders:

- CA2_L00203071_data_raw/ for all original unmodified datasets
- CA2_L00203071_data_processed/ for cleaned and transformed files, with filenames indicating the processing stage (e.g., hse_hospitals_cleaned.csv, nurses_per_county_transformed.csv)
- Google Colab: The analysis notebook will load CSV files directly from Google Drive using `pd.read_csv`

I will not upload anything with personal data, only public, cleaned data.

WHAT IS MY “PROBLEM”?

The main problem this report examines is the potential disconnect between where nurses are needed, where they are placed and where they can afford to live.

Based on my experience recruiting nurses from India, the Philippines, and Africa, I noticed a recurring theme: nurses recruited to fill vacancies in Irish hospitals struggled to secure accommodation in the areas where they were hired. Employers only provided one month of temporary housing upon arrival and after that, those nurses were left to search for a somewhere safe and close to where they worked in a rental market which was already under pressure from under supply and including an increased demand following the war in Ukraine.

This creates four interconnected problems:

1. **Affordability:** Nurses on starting HSE salaries (approx. € 34,000 to € 45,000) may be priced out of the market, especially in bigger cities.
2. **Hospital concentration:** Ireland’s major hospital are concentrated in a small number of urban areas. This forces nurses to live in places with higher living costs.
3. **Commuting:** Some nurses are forced to live far from their workplace in order to afford their rent. They face long commutes resulting in fatigue which will then impact their work.
4. **Retention:** The housing crisis may force nurses to leave the profession or emigrate to countries with more affordable housing.

METHOD

The analysis will be carried out entirely in Python using pandas, matplotlib, and seaborn within Google Colab.

Step 1: Data extraction from PDFs

The NMBI State of the Register 2025 is published as a PDF. I will manually transcribe the relevant tables into Excel and saved as CSV files. I attempted to use tabula-py to automate this extraction but found the output required more correction than manual entry at this stage of my learning. HSE salary scales will be extracted from PDF using Power Query and loaded onto an Excel worksheet.

Step 2: Data loading and inspection

Each dataset will be imported using `pd.read_csv()`. An initial inspection will be run on each file where relevant:

```
python
df.head()
df.shape()
df.info()
df.isnull().sum()
df.describe()
```

These checks establish dataset dimensions, data types, and missing values.

Step 3: Cleaning and structure setup

Unnecessary columns will be dropped using:

```
python
df.drop(columns=[...], axis=1, inplace=True)
```

All datasets will use a standardised county name column to enable merging.

Where needed, datasets will be aggregated at county level using:

```
python
df.groupby('County').size()
```

Step 4: Merging datasets

The cleaned datasets will be merged by 'County' into a single master dataframe:

```
python
master_df = pd.merge(population, nurses, on='County', how='left')
master_df = pd.merge(master_df, hospitals, on='County', how='left')
master_df = pd.merge(master_df, rent, on='County', how='left')
master_df = pd.merge(master_df, houseprice, on='County', how='left')
```

Step 5: Calculate derived metrics

Three key variables will be calculated to analyse trends and help answer questions like:

- Since the NMBI report does not break down nurses by country of qualification at county level, a proportional estimate will be applied.: Non-EU educated: 43,894 (48% of registered) (NMBI 2025) FIG. 1, PAGE 3.
- Hospital density per 10,000 people to compare demand across counties. Standard health metrics like WHO nurse density use the same "per 10,000" scaling (UNECE).
- Rental affordability ratio, rent as a proportion of gross nurse salary (€34,000 starting). Counties exceeding 0.30 (GRABENSTEIN, H., 2025) will be flagged as high cost areas for newly recruited nurses.

Step 6: Visualise

Charts and maps will be generated using matplotlib and seaborn, as outlined in the Analysis and Visualisation Plan below.

RELATED STUDIES

Previous studies confirm housing affordability blocks key workers like nurses, especially internationals, with geographic mismatches between jobs and liveable areas hitting recruitment hard. HUMPHRIES, N. ET AL. (2009) found migrant nurses in Ireland leave due to sky-high rents and commutes, often heading to Canada for cheaper options. Recent analysis shows our nurse shortages persist despite high numbers, as costs drive emigration O CHIMA, C. ET AL. (2026). Geographic supply gaps in healthcare further cluster services in pricey urban spots SMITH, S. ET AL. (2021). I'm building on this by mapping nurse numbers, hospitals, and county rents to spotlight Ireland's exact mismatches.

DATA CLEANING**Hospital Dataset (GeoHive)**

- Address fields were split inconsistently; a new 'FullAddress' field was created in Excel using concatenation and the =TRIM() formula to remove extra spaces.
- After sorting by Eircode, 29 Dublin entries included legacy area codes (e.g., "Dublin 4"); these were manually corrected.
- Duplicate entries for Temple Street Children's Hospital, The Mater, St. Vincent's University Hospital, and University Hospital Cork were confirmed via Google Maps/Eircode matching and removed.
- A new County column was derived using =TEXTAFTER() to extract county names from the end of address strings, ensuring consistent county identification.
- File saved as .csv for import into Python.

Nursing Dataset (NMBI)

The NMBI report contains multiple tables by nurse discipline, which I manually extracted and saved as CSV files. While several tables were relevant, the primary one for analysis is the Geographical distribution in Ireland of all practising registrants by county.

Housing Datasets (Property Price Register & RTB Rent Index)

- Unnecessary columns (e.g. date of sale, address, property type) will be removed.
- Average rent price and sale prices will be calculated at county level using:

```
python
df.groupby('County')['Price'].mean()
```

- The RTB Rent Index data will be cleaned to remove any incomplete county records and standardised to match county names used across all datasets

DATA TRANSFORMATION

Three new variables will be created from the merged dataset to support the analysis:

Estimated international nurses per county

python

```
master_df['Estimated Intl Nurses'] = master_df['Total Nurses'] * 0.48
```

Hospital density per 10,000 population

python

```
master_df['Hospitals per 10k'] = (master_df['Hospitals'] / master_df['Population']) * 10000
```

Rental affordability ratio

python

```
avg_nurse_salary = 34000
master_df['Rent % Salary'] = (master_df['Average Rent Price'] * 12) / avg_nurse_salary
```

MEASURE OF SUCCESS

Success will be judged by whether the datasets merge cleanly, the county metrics make sense, and the visualisations clearly show patterns in nurse demand and housing pressure.

DATA VISUALIZATIONS

The visualisation plan will use a horizontal bar chart, grouped bar chart, scatter plot, and maps to show rent, hospital/nurse density, and affordability.

Bar Charts

I will use the bar charts to rank counties by rent burden and nurse numbers and spot if places like Dublin high cost, maybe low demand, get too many international nurses. It gives clear visual proof of placement mismatches.

Scatter Plot

I will use a scatter plot to show hospital density (demand) vs rent % salary (cost) to show if high-need areas are unaffordable or nurses end up concentrated in low-demand, expensive counties.

Maps

Maps would be perfect for this analysis, especially choropleth and heatmaps, they'd colour Ireland's counties by nurse density, rent costs, or hospital needs, instantly showing geographic patterns like Dublin overload vs rural gaps. However, as I have no prior experience with geospatial plotting in Python (like geopandas or folium), I'll research if it's practical during the project and use charts as backup if needed.

RESULTS AND IMPACT ANALYSIS

Expected results:

I expect to find that counties with the most hospitals (Dublin, Cork, Galway) also have the highest rents and house prices. I expect international nurses to be concentrated in those same counties. This would mean newly arrived nurses face the worst affordability just when they are starting their careers.

Impact:

If my analysis shows a clear mismatch, then:

- Employers could offer more than one month of temporary housing.
- The government could look at key worker housing for nurses in high cost areas.
- Recruitment agencies could give better advice about living costs before nurses move to Ireland.

References

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